The inclusive **or** is only wrong if both propositions are false. For example,

Albert Einstein discovered America [4] ***or*** *Isaac Newton was Russian.*

The inclusive **o**r has a set theoretic interpretation as the union of two sets: it denotes the subset containing anything in either or both of the component subsets. In the die example, (*A* **or** *B*) denotes the subset {1, 2, 3, 5}.

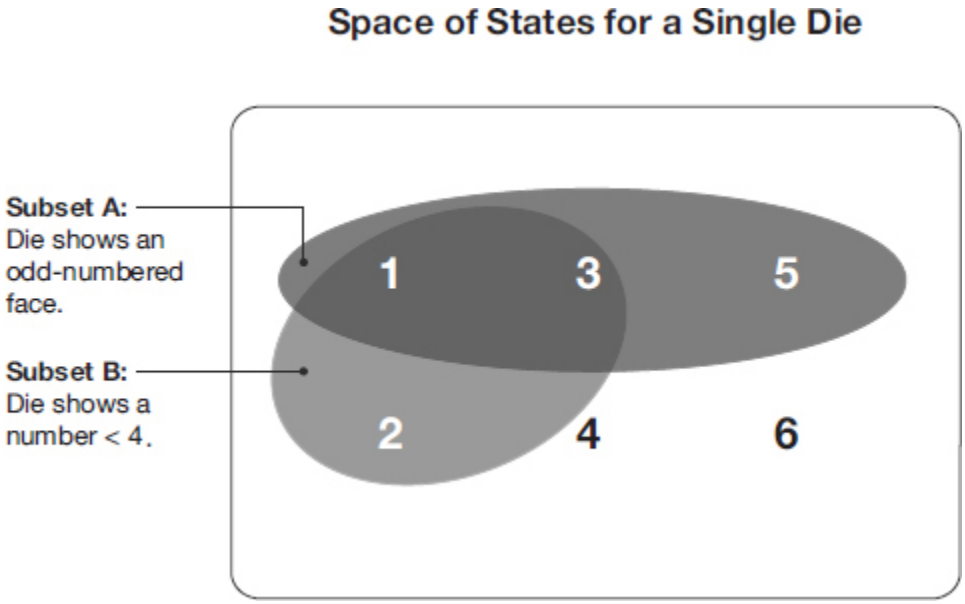


Figure 1.5: An Example of the Classical model of State Space. Subset *A* represents the proposition "the die shows an odd-numbered face." Subset *B*: "The die shows a number < 4." Dark shading shows the intersection of *A* and *B*, which represents the proposition (*A* **and** *B*). White numbers are elements of the union of *A* with *B*, representing the proposition (*A* **or** *B*).

1.6 TESTING CLASSICAL PROPOSITIONS

Let's return to the simple quantum system consisting of a single spin, and the various propositions whose truth we could test using the apparatus $\mathcal{A}$. Consider the following two propositions:

A: The *z* component of the spin is +1.

B: The *x* component of the spin is +1.